
LABORATORY MANUAL FOR MATHEMATICS PRACTICALS (WITH FOSS TOOLS)

Discipline Specific Core Courses(DSCC)

For 2nd Semester
B.Sc. (Revised NEP)

**C2MAT1P1–Practicals on
Number Theory and Calculus – II
FOSS - SciLab**



**K.L.E. SOCIETY's
S.K. ARTS COLLEGE AND H.S.K.
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Affiliated to Karnatak University, Dharwad

**Effective from
the academic year(2024-25)**

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* Introduction to the Commands Related to the experiments (SciLab)

modulo() function:

`i = modulo(n,m)`

`modulo` computes $i = n \pmod{m}$ this means i is the remainder of “ n divided by m ”.

⇒ $i = n - m * \text{int}(n/m)$. Here the answer may be negative if n or m are negative.

Examples :

```
-->modulo(5,3)
```

```
ans =  
  2.
```

```
-->modulo(7,3)
```

```
ans =  
  1.
```

```
-->modulo(7,4)
```

```
ans =  
  3.
```

```
-->modulo(6,2)
```

```
ans =  
  0.
```

factor function:

`factor(x)`

Given a real x , `factor(x)` returns in a vector y the primes numbers decomposition of x .

Particular cases: `factor(0)` returns 0, and `factor(1)` returns 1

Examples:

```
-->factor(221)
```

```
ans =  
 13. 17.
```

```
-->factor(21)
```

```
ans =  
  3.  7.
```

```
-->factor(15)
```

```
ans =  
  3.  5
```

gcd() function:

Greatest Common Divisor (or Highest Common Factor)

u = gcd(p)

where p is a row vector $p = [p_1, \dots, p_n]$

u is greatest common divisor of a row vector p.

Examples:

```
-->gcd([2,3])
```

```
ans =
```

```
1.
```

```
-->gcd([12,13])
```

```
ans =
```

```
1.
```

```
-->gcd([4,6])
```

```
ans =
```

```
2.
```

factorial function :

The factorial function

```
f = factorial( n )
```

Returns the factorial of n, that is, the product of all integers $1 * 2 * \dots * n$.

Note: This function overflows as soon as $n > 170$.

```
-->factorial(4)
```

```
ans =
```

```
24
```

```
// See the limits of factorial: f(171)=%inf
```

```
factorial(170) // 7.257415615307998967e306
```

```
factorial(171) // %inf
```

numderivative() function :

Approximate derivatives of a function

```
f1=numderivative(f,x)
```

```
[f1,f2]=numderivative(f,x)
```

Where f- a function or a list , the function to differentiate

x- a 'n by 1' or '1 by n' vector of doubles, real, the point where to compute the derivative

f1- first derivative of function

f2- second derivative of function

Examples:

```
function y=f(x)
```

```
y=x^3
```

```
endfunction
```

```
x=1
```

```
y1=numderivative(f,x)
```

```
disp("derivative of 1st =" + string(y1))
```

```
[y1,y2]=numderivative(f,x)
```

```
disp("derivtive of 1st= " + string(y1))
```

```
disp("derivative of 2nd = " + string(y2))
```

output

```
"derivative of 1st =3"
```

```
"derivative of 1st= 3"
```

```
"derivative of 2nd = 6"
```

integrate() function :

```
y = integrate(expr, v, x0, x1)
```

where

expr - a character string defining a Scilab expression.

v - a character string, the integration variable name.

x0 - a real number, the lower bound of integration.

x1 - a vector of real numbers, upper bounds of integration

Examples

```
--> integrate('x^2','x',0,1)
```

```
ans =
```

```
0.3333333
```

```
--> integrate('sin(x)','x',0,%pi/2)
```

```
ans =
```

```
1.0000000
```

```
--> integrate('cos(x)','x',0,%pi/2)
```

```
ans =
```

```
1.0000000
```

LAB 1: Euler's Theorem for number theory

Program to solve examples using Euler's theorem

//Program to solve examples using Euler's theorem

```
clc
function phi=euler_p(n)
phi=0
for i=1:n-1
if gcd([i,n])==1 then
phi=phi+1
end
end
endfunction
n=input("enter the value of n ")
b=input("enter the base value of b ")
c=input("enter the power value of c ")
pn=euler_p(n)
disp("Phi of n is "+string(pn))
m=int(c/pn)
k=c-m*pn
if c>=pn then
if gcd([b,n])==1 then
r=modulo((b^pn)^m,n)
if k~=0 then
r=modulo(b^k,n)
end
disp("by euler's theorem, remainder/last digit is "+string(r))
else
disp("eulers theorem not applicable")
end
else
disp("power is less then phi of n")
end
end
```

OUTPUT

enter the value of n 7
enter the base value of b 3
enter the power value of c 50
Phi of n is 6
by euler's theorem, remainder/last digit is 2

OUTPUT

enter the value of n 10
enter the base value of b 9
enter the power value of c 51
Phi of n is 4
by euler's theorem, remainder/last digit is 9

Example: Find the remainder when 7^{102} is divided by 12

**Solution: Here $n=12$, $b=7$, $c=102$, $(7,12)=1$
 $\Phi(12)=4$ and remainder is 1**

OUTPUT

enter the value of n 12
enter the base value of b 7
enter the power value of c 102
Phi of n is 4
by euler's theorem, remainder/last digit is 1

Example: Find the remainder when 3^{50} is divided by 7

**Solution: Here $n=7$, $b=3$, $c=50$, $(3,7)=1$
 $\Phi(7)=6$ and remainder is 2**

Example: Find the last digit of 9^{51}

**Solution: Here $n=10$, $b=9$, $c=51$, $(9,10)=1$
 $\Phi(10)=4$ and last digit is 9**

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LAB 2: Program to solve examples using Fermat's theorem

//Program to solve examples using Fermat's theorem

```
clc
function pr=fermat_p(n)
pr=%T
for i=2:sqrt(n)
if modulo(n,i)==0 then
pr=%F
break
end
end
endfunction
n=input("enter the value of n ")
b=input("enter the base value of b ")
c=input("enter the power value of c ")
pp=fermat_p(n)
if pp==%T then
pn=n-1
m=int(c/pn)
k=c-m*pn
if c>=pn then
if gcd([b,n])==1 then
r=modulo((b^pn)^m,n)
if k~=0 then
r=modulo(b^k,n)
end
disp("by fermats theorem,remainder is "+string(r))
else
disp("fermats theorem is not applicable")
end
else
disp("power is less than n-1")
end
else
disp("given number is not a prime")
end
```

OUTPUT

enter the value of n 7
enter the base value of b 3
enter the power value of c 50
by fermats theorem,remainder is 2

Example: Find the remainder when 3^{50} is divided by 7
Solution: Here $n=7$, $b=3$, $c=50$, $(3, 7)=1$ n is prime number and remainder is 2

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LAB 03: Program to solve examples using Wilson's theorem

//Program to solve examples using Wilson's theorem

```
clc
n=input("enter the value of n ")
d=input("enter the value of d ")
p=input("enter the value of p ")
pr=factor(p)
disp("prime factors of "+string(p))
disp(pr)
k=size(pr)
r=1
for i=1:k(2)
m=modulo(factorial(n)+d,pr(i))
disp("remainder when divided by "+string(pr(i))+" is "+string(m))
r=r*m
end
if r==0 then
disp(string(n)+"!+"string(d)+" is divisible by "+string(p))
else
disp(string(n)+"!+"string(d)+" is not divisible by "+string(p))
end
```

OUTPUT

enter the value of n 18
enter the value of d 1
enter the value of p 437
prime factors of 437
19. 23.

remainder when divided by 19 is 0
remainder when divided by 23 is 0
18!+1 is divisible by 437

Example: Prove that 18! +1 is divisible by 437 by Wilson's theorem

Solution: Here n=18, p=437, d=1,
prime factors of p are 19 & 23
18! +1 is divisible by 437

OUTPUT

enter the value of n 12
enter the value of d 209
enter the value of p 221
prime factors of 221
13. 17.

remainder when divided by 13 is 0
remainder when divided by 17 is 0
12!+209 is divisible by 221

Example: Prove that 12! +209 is divisible by 221

Solution: Here n=12, p=221, d=209,
prime factors of p are 13 & 17
12! +209 is divisible by 221

LAB 04: Program on reduction formulas for $\int \sin^n(x) dx$ (for all positive values of n) with suitable examples.

$$\int \sin^n(x) dx = -\frac{\sin^{n-1}(x)\cos(x)}{n} + \frac{(n-1)}{n} \int \sin^{n-2}(x) dx$$

//Program on reduction formulas for $\int \sin^n(x) dx$ (for all positive values of n)

```
clc
function I_n=sin_reduction(n)
if n==0 then
I_n="x"
elseif n==1 then
I_n="-cos(x)"
else
s1=l+"-1/"+string(n)+r+"cos(x)sin(x)"+string(n-1)
t=l+string((n-1))+"/"+string(n)+r
s2=t+l+sin_reduction(n-2)+r
I_n=s1+"+"+s2
end
end
endfunction
l="(" //left bracket
r=")" //right bracket
n=input("Enter the value of n ")
result=sin_reduction(n)
disp("The value of  $\int \sin^n(x) dx$  is")
disp(result)
```

OUTPUT

enter the value of n 3
The value of $\int \sin^3(x) dx$ is
 $(-1/3)\cos(x)\sin(x)^2+(2/3)(-\cos(x))$

Example: Evaluate $\int \sin^3 x dx$

Solution: Here n=3

$$\int \sin^3 x dx = -\frac{1}{3} \cos(x) \sin^2(x) + \frac{2}{3} (-\cos(x))$$

OUTPUT

enter the value of n 4
The value of $\int \sin^4(x) dx$ is
 $(-1/4)\cos(x)\sin(x)^3+(3/4)((-1/2)\cos(x)\sin(x)^1+(1/2)(x))$

Example: Evaluate $\int \sin^4 x dx$

Solution: Here n=4

$$\begin{aligned} \int \sin^4 x dx \\ = -\frac{1}{4} \cos(x) \sin^3(x) + \frac{3}{4} \left[-\frac{1}{2} \cos(x) \sin(x) + \frac{1}{2} x \right] \end{aligned}$$

LAB 04: Program on reduction formulas for $\int \cos^n(x) dx$ (for all positive values of n) with suitable examples

$$\int \cos^n(x) dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x dx$$

//Program on reduction formulas for $\int \cos^n(x) dx$ (for all positive values of n)

clc

function I_n=cos_reduction(n)

if n==0 then

I_n="x"

elseif n==1 then

I_n="sin(x)"

else

s2=l+"1/"+string(n)+r+"sin(x)cos(x)^"+string(n-1)

t=l+string(n-1)+"/"+string(n)+r

s1=l+cos_reduction(n-2)+r

I_n=s2+" "+t+s1

end

endfunction

l="(" //left bracket

r=")" //right bracket

n=input("enter the power n ")

result=cos_reduction(n)

disp("the value of $\int \cos^n(x) dx$ is")

disp(result)

OUTPUT

enter the power n 3

The value of $\int \cos^3(x) dx$ is
 $(1/3)\sin(x)\cos(x)^2 + (2/3)(\sin(x))$

Example: Evaluate $\int \cos^3 x dx$

Solution: Here n=3

$$\int \cos^3 x dx = \frac{1}{3} \sin(x) \cos^2(x) + \frac{2}{3} \sin(x)$$

OUTPUT

enter the power n 4

the value of $\int \cos^4(x) dx$ is
 $(1/4)\sin(x)\cos(x)^3 + (3/4)((1/2)\sin(x)\cos(x)^1 + (1/2)(x))$

Example: Evaluate $\int \cos^4 x dx$

Solution: Here n=4

$$\begin{aligned} \int \cos^4 x dx \\ = \frac{1}{4} \sin(x) \cos^3(x) + \frac{3}{4} \left[\frac{1}{2} \sin(x) \cos(x) + \frac{1}{2} x \right] \end{aligned}$$

LAB 05: Program on reduction formulas for $\int_0^{\frac{\pi}{2}} \sin^n(x) dx$ (for all positive values of n) with suitable examples

$$\int_0^{\frac{\pi}{2}} \sin^n(x) dx = \frac{n-1}{n} \times \frac{n-3}{n-2} \times \frac{n-5}{n-4} \times \dots \begin{cases} \frac{1}{2} \times \frac{\pi}{2} & \text{if } n \text{ is even} \\ \frac{2}{3} & \text{if } n \text{ is odd} \end{cases}$$

```
//Program on reduction formulas for [sin^n(x)dx](for all positive values of n) with the limit 0 to pi/2
clc
p=1
n=input("enter the value of n ")
for i=1:2:n-1
p=p*((n-i)/(n-i+1))
end
if modulo(n,2)==0 then
p=p*%pi/2
end
disp("The value of integral sin(x)^(n) with limit 0 to pi/2 is")
disp(p)
disp("using built function the value of integral is")
disp(integrate('sin(x)^n','x',0,%pi/2))
```

OUTPUT

enter the value of n 5
The value of integral sin(x)^5 with limit 0 to pi/2 is
0.5333333
using built function the value of integral is
0.5333333

Example: Evaluate $\int_0^{\frac{\pi}{2}} \sin^5(x) dx$

Solution: Here n=5

$$\int_0^{\frac{\pi}{2}} \sin^5(x) dx = \frac{4}{5} \times \frac{2}{3} = \frac{8}{15} = 0.53333$$

OUTPUT

enter the value of n 6
The value of integral sin(x)^6 with limit 0 to pi/2 is
0.4908739
using built function the value of integral is
0.4908739

Example: Evaluate $\int_0^{\frac{\pi}{2}} \sin^6(x) dx$

Solution: Here n=6

$$\int_0^{\frac{\pi}{2}} \sin^6(x) dx = \frac{5}{6} \times \frac{3}{2} \times \frac{1}{2} \times \frac{\pi}{2} = \frac{5\pi}{32} = 0.49087$$

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LAB 05: Program on reduction formulas for $\int_0^{\frac{\pi}{2}} \cos^n(x) dx$ (for all positive values of n) with suitable examples.

$$\int_0^{\frac{\pi}{2}} \cos^n(x) dx = \frac{n-1}{n} \times \frac{n-3}{n-2} \times \frac{n-5}{n-4} \times \dots \begin{cases} \frac{1}{2} \times \frac{\pi}{2} & \text{if } n \text{ is even} \\ \frac{2}{3} & \text{if } n \text{ is odd} \end{cases}$$

//Program on reduction formulas for $\int \cos^n(x) dx$ (for all positive values of n) with the limit 0 to $\pi/2$

`clc`

`p=1`

`n=input('enter the value of n ')`

`for i=1:2:n-1`

`p=p*((n-i)/(n-i+1))`

`end`

`if modulo(n,2)==0 then`

`p=p*%pi/2`

`end`

`disp('the value of integral cos(x)^(n) with limit 0 to pi/2 is ')`

`disp(p)`

`disp('using built function the value of integral is')`

`disp(integrate('cos(x)^n','x',0,%pi/2))`

OUTPUT

enter the value of n 6

the value of integral cos(x)^6 with limit 0 to pi/2 is

0.4908739

using built function the value of integral is

0.4908739

Example: Evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x) dx$

Solution: Here n=6

$$\int_0^{\frac{\pi}{2}} \cos^6(x) dx = \frac{5}{6} \times \frac{3}{2} \times \frac{1}{2} \times \frac{\pi}{2} = \frac{5\pi}{32} = 0.49087$$

OUTPUT

enter the value of n 5

the value of integral cos(x)^5 with limit 0 to pi/2 is

0.5333333

using built function the value of integral is

0.5333333

Example: Evaluate $\int_0^{\frac{\pi}{2}} \cos^5(x) dx$

Solution: Here n=5

$$\int_0^{\frac{\pi}{2}} \cos^5(x) dx = \frac{4}{5} \times \frac{2}{3} = \frac{8}{15} = 0.53333$$

LAB 06: Program on reduction formulas for $\int \tan^n(x) dx$ (for all positive values of n) with suitable examples

$$\int \tan^n(x) dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x dx$$

//Program on reduction formulas for $\int \tan^n(x) dx$ (for all positive values of n)

```
clc
function I_n=tan_reduction(n)
if n==0 then
I_n="x"
elseif n==1 then
I_n="-log|cos(x)|"
else
s2=l+"1/"+string(n-1)+r+"tan(x)^(n-1)+string(n-1)
s1="(-1)"+l+tan_reduction(n-2)+r
I_n=s2+"+"+s1
end
end
endfunction
l="(" //left bracket
r=")" //right bracket
n=input("enter the value of n ")
result=tan_reduction(n)
disp("The value of integral tan(x)^(n) is ")
disp(result)
```

OUTPUT

enter the value of n 5
The value of integral tan(x)^5 is
(1/4)tan(x)^4+(-1)((1/2)tan(x)^2+(-1)(-log|cos(x)|))

Example: Evaluate $\int \tan^5(x) dx$

Solution: Here n=5

$$\int \tan^5(x) dx = \frac{1}{4} \tan^4(x) - \left[\frac{1}{2} \tan^2(x) - (-\log|\cos(x)|) \right]$$



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LAB 06: Program on reduction formulas for $\int \cot^n(x) dx$ (for all positive values of n) with suitable examples

$$\int \cot^n(x) dx = -\frac{\cot^{n-1} x}{n-1} - \int \cot^{n-2} x dx$$

//Program on reduction formulas for $\int \cot^n(x) dx$ (for all positive values of n)

```
clc
function I_n=cot_reduction(n)
if n==0 then
I_n='x'
elseif n==1 then
I_n='log|sin(x)|'
else
s1=l+'-1/'+string(n-1)+r+'cot(x)'+string(n-1)
s2='(-1)'+l+cot_reduction(n-2)+r
I_n=s1+'+'+s2
end
end
endfunction
l='(' //left bracket
r=')' //right bracket
n=input('enter the value of n')
result=cot_reduction(n)
disp('The value of integral is')
disp(result)
```

OUTPUT

```
enter the value of n 4
The value of integral is
(-1/3)cot(x)^3+(-1)((-1/1)cot(x)^1+(-1)(x))
```

Example: Evaluate $\int \cot^4(x) dx$

Solution: Here $n=4$

$$\int \cot^4(x) dx = -\frac{1}{3} \cot^3(x) - [-\cot(x) - x]$$

LAB 07: Program on reduction formulas for $\int \sec^n(x) dx$ (for all positive values of n) with suitable examples

$$\int \sec^n x dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x dx$$

//Program on reduction formulas for $\int \sec^n(x) dx$ (for all positive values of n)

```
clc
function I_n=sec_reduction(n)
if n==0 then
I_n='x'
elseif n==1 then
I_n='log|sec(x)+tan(x)|'
elseif n==2 then
I_n='tan(x)'
else
t=l+'1/'+string(n-1)+r
s1=l+'tan(x)sec(x)'+string(n-2)+r
s2=l+string(n-2)+'/'+string(n-1)+r+l+sec_reduction(n-2)+r
I_n=t+s1+'+'+s2
end
end
endfunction
l='(' //left bracket
r=')' //right bracket
n=input('enter the value of n ')
result=sec_reduction(n)
disp('the value of integral is')
disp(result)
```

OUTPUT

enter the value of n 5

the value of integral is

$(1/4)(\tan(x)\sec(x)^3) + (3/4)((1/2)(\tan(x)\sec(x)^1) + (1/2)(\log|\sec(x) + \tan(x)|))$

Example: Evaluate $\int \sec^5(x) dx$

Solution: Here n=5

$$\int \sec^5(x) dx = \frac{1}{4} \tan(x) \sec^3(x) + \frac{3}{4} \left[\frac{1}{2} \tan(x) \sec(x) + \frac{1}{2} \log|\sec(x) + \tan(x)| \right]$$

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LAB 07: Program on reduction formulas for $\int \operatorname{cosec}^n(x) dx$ (for all positive values of n) with suitable examples

$$\int \operatorname{cosec}^n x \, dx = -\frac{\operatorname{cosec}^{n-2} x \cot x}{n-1} - \frac{n-2}{n-1} \int \operatorname{cosec}^{n-2} x \, dx$$

//Program on reduction formulas for $\int \operatorname{cosec}^n(x) dx$ (for all positive values of n)

clc

function I_n=cosec_reduction(n)

if n==0 then

I_n='x'

elseif n==1 then

I_n='log|cosec(x)-cot(x)|'

elseif n==2

I_n='-cot(x)'

else

t=l+'-1/'+string((n-1))+r

s1='cot(x)cosec(x)'+string(n-2)

s2=l+string(n-2)+'/'+string(n-1)+r+l+cosec_reduction(n-2)+r

I_n=t+s1+'+'+s2

end

end

endfunction

l='(' //left bracket

r=')' //right bracket

n=input('enter the value of n ')

result=cosec_reduction(n)

disp('the value of integral is')

disp(result)

OUTPUT

enter the value of n 6

the value of integral is

$(-1/5)\cot(x)\operatorname{cosec}(x)^4 + (4/5)((-1/3)\cot(x)\operatorname{cosec}(x)^2 + (2/3)(-\cot(x)))$

Example: Evaluate $\int \operatorname{cosec}^6(x) dx$

Solution: Here $n=6$

$$\begin{aligned} & \int \operatorname{cosec}^6(x) dx \\ &= -\frac{1}{5} \cot(x) \operatorname{cosec}^4(x) + \frac{4}{5} \left[-\frac{1}{3} \cot(x) \operatorname{cosec}^2(x) + \frac{2}{3} (-\cot(x)) \right] \end{aligned}$$

LAB 08: Program on reduction formulas for $\int x^m (\log x)^n dx$ (for all positive values of n,m) with suitable examples

$$\int x^m (\log x)^n dx = \frac{x^{m+1} (\log x)^n}{m+1} - \frac{n}{m+1} \int x^m (\log x)^{n-1} dx$$

//Program on reduction formulas for $\int x^m \log x^n(x) dx$ (for all positive values of n,m)

clc

function result=xlog_reduction(m, n)

if n==0 then

result=l+"1/"+string(m+1)+r+l+"x^"+string(m+1)+r

else

s1=l+"1/"+string(m+1)+r+l+"x^"+string(m+1)+" log(x)^"+string(n)+r

s2=l+string(n)+" / "+string(m+1)+r+l+xlog_reduction(m,n-1)+r

result=s1+"-"+s2

end

endfunction

l="(" //left bracket

r=")" //right bracket

m=input("enter the value of m ")

n=input("enter the value of n ")

result=xlog_reduction(m,n)

disp("the value of integral is")

disp(result)

OUTPUT

enter the value of m 3

enter the value of n 3

the value of integral is

$(1/4)(x^4 \log(x)^3) - (3/4)((1/4)(x^4 \log(x)^2) - (2/4)((1/4)(x^4 \log(x)^1) - (1/4)((1/4)(x^4))))$

Example: Evaluate $\int x^3 (\log x)^3 dx$

Solution: Here m=3, n=3

$$\begin{aligned} & \int x^3 (\log x)^3 dx \\ &= \frac{1}{4} x^4 (\log x)^3 - \frac{3}{4} \left[\frac{1}{4} x^4 (\log x)^2 - \frac{2}{4} \left\{ \frac{1}{4} x^4 \log x - \frac{1}{4} \left(\frac{1}{4} x^4 \right) \right\} \right] \end{aligned}$$

LAB 08: Program on reduction formulas for $\int x^n e^{ax} dx$ (for all positive values of n) with suitable examples

$$\int x^n e^{ax} dx = \frac{x^n e^{ax}}{a} - \frac{n}{a} \int x^{n-1} e^{ax} dx$$

//Program on reduction formulas for $\int x^n e^{(ax)} dx$ (for all positive values of n)

clc

function result=xexp_reduction(n, a)

if n==0 then

result=l+"1/"+string(a)+r+"e^"+l+string(a)+"x"+r

else

s1=l+"1/"+string(a)+r+l+"x^"+string(n)+" e^"+l+string(a)+"x"+r+r

s2=l+string(n)+"/"+string(a)+r+l+xexp_reduction(n-1,a)+r

result=s1+"-"+s2

end

endfunction

l="(" //left bracket

r=")" //right bracket

n=input("enter the value of n ")

a=input("enter the value of a ")

result=xexp_reduction(n,a)

disp("the value of integral is")

disp(result)

OUTPUT

enter the value of n 2

enter the value of a 3

the value of integral is

(1/3)(x^2 e^(3x))-(2/3)((1/3)(x^1 e^(3x))-(1/3)((1/3)e^(3x)))

Example: Evaluate $\int x^2 e^{3x} dx$

Solution: Here n=2, a=3

$$\int x^2 e^{3x} dx = \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \left[\frac{1}{3} x e^{3x} - \frac{1}{3} \left(\frac{1}{3} e^{3x} \right) \right]$$

LAB 09: Reduction of $\int \sin^m x \cos^n x dx$

Program on reduction formulas for $\int \sin^m x \cos^n x dx$ (for all positive values of m and n) with suitable examples

$$\int \sin^m x \cos^n x dx = \frac{-\cos^{n+1} x \sin^{m-1} x}{m+n} + \frac{m-1}{m+n} \int \sin^{m-2} x \cos^n x dx$$

//Program on reduction formulas for $\int \sin^m(x) \cos^n(x) dx$ (for all positive values of n)

//by decomposition of sine function

clc

function result=sin_cos_reduction(m,n)

if n==0 then

if m==0 then

result="x"

elseif m==1 then

result="(-cos(x))"

end

elseif m==0 then

if n==1 then

result="sin(x)"

else

s=sin_cos_reduction(0,n-2)

result="(1/"+string(n)+")sin(x) cos(x)^"+string(n-1)+" (" +
string(n-1)+"/"+string(n)+")("+s+")"

end

elseif m>1 then

s=sin_cos_reduction(m-2,n)

result="(-1/"+string(m+n)+")sin(x)^"+string(m-1)+" cos(x)^"+
string(n+1)+")"+" (" +string(m-1)+"/"+string(m+n)+")("+s+")"

else

result="(-1/"+string(m+n)+")cos(x)^"+string(n+1)

end

endfunction

m=input("enter the sine power m ")

n=input("enter the cosine power n ")

res=sin_cos_reduction(m,n)

disp("the value of integral is")

disp(res)

Output

enter the sine power m 3

enter the cosine power n 2

the value of integral is

$(-1/5)(\sin(x)^2 \cos(x)^3) + (2/5)((-1/3)\cos(x)^3)$

Output

enter the sine power m 4

enter the cosine power n 3

the value of integral is

$(-1/7)(\sin(x)^3 \cos(x)^4) + (3/7)((-1/5)(\sin(x)^1 \cos(x)^4) + (1/5)((1/3)\sin(x) \cos(x)^2 + (2/3)(\sin(x))))$

Example: Evaluate $\int \sin^3 x \cos^2 x dx$

Solution: Here m=3, n=2

$$\int \sin^3 x \cos^2 x dx = -\frac{1}{5} \cos^3(x) \sin^2(x) - \frac{2}{5} \cos^3(x)$$

Example: Evaluate $\int \sin^4 x \cos^3 x dx$

Solution: Here m=4, n=3

$$\int \sin^4 x \cos^3 x dx = -\frac{1}{7} \sin^3(x) \cos^4(x) + \frac{3}{7} \left[-\frac{1}{5} \sin(x) \cos^4(x) + \frac{1}{5} \left\{ \frac{1}{3} \sin(x) \cos^2(x) + \frac{2}{3} \sin(x) \right\} \right]$$

LAB 09: Program on reduction formulas for $\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx$ (for all positive values of m and n) with suitable examples

/Program on reduction formulas for $\int \sin^m(x) \cos^n(x) dx$ with limit 0 to $\pi/2$ (for all positive values of m and n)

clc

I='sin(x)^m*cos(x)^n'

m=input("enter the sine power ")

n=input("enter the cosine power ")

y=integrate(I,'x',0,%pi/2)

disp("the value of the integral sin(x)^"+string(m)+" cos(x)^"+string(n)+" is "+string(y))

Output

enter the sine power m 3
enter the cosine power n 4

"the value of the integral sin(x)^3 cos(x)^4 is 0.0571429"

Example: Evaluate $\int \sin^3 x \cos^4 x dx$

Solution: Here m=3, n=4

$$\int_0^{\frac{\pi}{2}} \sin^3 x \cos^4 x dx = \frac{2}{35} = 0.05714$$



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LAB 10: Euler's Theorem on homogeneous functions

Program to verify the Euler's theorem and its extension

//Program to verify the Euler's theorem and its extension

clc

function u=f(x)

u=x(1)^2+x(2)^2-2*x(1)*x(2)

endfunction

n=2 *//order of homogeneous function*

x=[2,1]

[fu,su]=numderivative(f,x)

disp("First Partial derivative of u (ux and uy) are")

disp(fu)

disp("second Partial derivative of u (uxx,uxy,uyx and uyy) are")

disp(su)

L=x(1)*fu(1)+x(2)*fu(2)

R=n*f(x)

disp("x ux + y uy = "+string(L)+" and nu = "+string(R))

if abs(R-L)<=0.00001 **then**

disp("Eulers Theorem is verified")

else

disp("Eulers Theorem is not verified")

end

L=x(1)^2*su(1)+2*x(1)*x(2)*su(2)+x(2)^2*su(4)

R=n*(n-1)*f(x)

disp("x^2 uxx+2xy uxy+y^2 uyy = "+string(L)+" and n(n-1)u = "+string(R))

if abs(R-L)<=0.00001 **then**

disp("Eulers Extension Theorem is verified")

else

disp("Eulers extension Theorem is not verified")

end

Output

First Partial derivative of u (ux and uy) are

2. -2.

Second Partial derivative of u (uxx,uxy,uyx and uyy) are

2. -2. -2. 2.

x ux + y uy = 2 and nu = 2

Eulers Theorem is verified

x^2 uxx+2xy uxy+y^2 uyy = 2 and n(n-1)u = 2

Eulers Extension Theorem is verified

Example: Verify Eulers Theorem for the function $u = x^2 + y^2 - 2xy$

Solution: $u_x = 2x - 2y$, $u_y = 2y - 2x$

$$u_{xx} = 2, u_{xy} = -2, u_{yx} = -2, u_{yy} = 2$$

At (2,1)

$$u_x = 2, u_y = -2, u_{xx} = 2, u_{xy} = -2$$

$$u_{yx} = -2, u_{yy} = 2$$

Thus $x u_x + y u_y = 2u$ and

$$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = n(n-1)u$$

LAB 11: Maclaurin's expansion

Programs to construct series using Maclaurin's expansion for functions of two variables

//Programs to construct series using Maclaurin's expansion for functions of two variables

clc

function u=f(p)

u=exp(p(1))*cos(p(2))

endfunction

p=[0,0]

[fu,su]=numderivative(f,p)

disp("First Partial derivative of u->ux and uy are")

disp(fu)

disp("second Partial derivative of u->uxx,uxy,uyx and uyy are")

disp(su)

t=""

if f(p)~=0 then

t=string(f(p))+""

end

if fu(1)~=0 then

t=t+(" "+string(fu(1))+"")x+"

end

if fu(2)~=0 then

t=t+(" "+string(fu(2))+"")y+"

end

if su(1)~=0 then

t=t+(" "+string(su(1))+"/2)x^2+"

end

if su(2)~=0 then

t=t+(" "+string(su(2))+"")xy+"

end

if su(4)~=0 then

t=t+(" "+string(su(4))+"/2)y^2+"

end

disp("the expansion of function using Maclaurin series upto 2nd degree is ")

disp("e^x cos(y) = "+t+"...")

OR

//Programs to construct series using Maclaurin's expansion for functions of two variables

clc

function u=f(p)

u=exp(p(1))*cos(p(2))

endfunction

p=[0,0]

[fu,su]=numderivative(f,p)

disp("First Partial derivative of u->ux and uy are")

disp(fu)

disp("second Partial derivative of u->uxx,uxy,uyx and uyy are")

disp(su)

t1=string(f(p))+""+"(" "+string(fu(1))+"")x+"(" "+string(fu(2))+"")y+"

t2="(" "+string(su(1))+"/2)x^2+"(" "+string(su(2))+"")xy+"(" "+string(su(4))+"")y^2+"

disp("the expansion of function using Maclaurin series upto 2nd degree is ")

disp("e^x cos(y) = "+t1+t2+"...")

Output

First Partial derivative of u->ux and uy are

1. 0.

second Partial derivative of u->uxx,uxy,uyx and uyy are

1. 0. 0. -1.

the expansion of function using Maclaurin series upto 2nd degree is

$e^x \cos(y) = 1 + (1)x + (1/2)x^2 + (-1/2)y^2 + \dots$

Expand the function $f(x, y) = e^x \cos y$ using maclaurin's expansion

$$f_x = e^x \cos y \quad f_y = -e^x \sin y$$

$$f_{xx} = e^x \cos y \quad f_{yy} = -e^x \cos y$$

$$f_{xy} = -e^x \sin y \quad f_{yx} = -e^x \sin y$$

$$\text{At } (0, 0) \quad f_x(0, 0) = 1 \quad f_y(0, 0) = 0 \quad f_{xx}(0, 0) = 1$$

$$f_{yy}(0, 0) = -1 \quad f_{xy}(0, 0) = 0 \quad f_{yx}(0, 0) = 0$$

$$f(x, y) = 1 + x + \frac{x^2}{2!} - \frac{y^2}{2!} + \dots$$

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LAB 12: Program to find gradient, divergence and curl

Gradient of Scalar Function F , $\text{grad}F = \nabla F = \hat{i} \frac{\partial F}{\partial x} + \hat{j} \frac{\partial F}{\partial y} + \hat{k} \frac{\partial F}{\partial z}$

//Program to find gradient

clc

function F=fun(x)

F=x(1)^2*x(2)^2*x(3)^2

endfunction

disp("Scalar function F= x^2*y^2*z^2")

x=[1,1,2]

f=numderivative(fun,x)

disp("grad of F is "+string(f(1))+"i "+string(f(2))+"j "+string(f(3))+"k")

OUTPUT

Scalar function F= x^2*y^2*z^2
grad of F is 8i+8j+4k

Find the gradient of a scalar function

$$F = x^2 y^2 z^2$$

$$\text{Solution: } F_x = 2xy^2z^2 \quad F_y = 2x^2yz^2$$

$$F_z = 2x^2y^2z$$

$$\nabla F = 2xy^2z^2\hat{i} + 2x^2yz^2\hat{j} + 2x^2y^2z\hat{k}$$

$$\text{At } (1,1,2) \quad \nabla F = 8\hat{i} + 8\hat{j} + 4\hat{k}$$

Divergence of the vector function $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$,

$$\text{div}\vec{F} = \nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

//Program to find divergence

clc

function F=fun(x)

F(1)=x(1)^2*x(2)

F(2)=x(2)*x(1)

F(3)=x(2)^2*x(3)

endfunction

disp("Vector function F=x^2y i+xy j+x^2z k")

x=[-1,2,3]

f=numderivative(fun,x)

disp("Partial derivatives computation are $\partial F_1/\partial x, \partial F_1/\partial y, \partial F_1/\partial z$ etc")

disp(f)

//div Fis $\partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$

df=f(1,1)+f(2,2)+f(3,3)

disp("div of F="+string(df))

OUTPUT

Vector function F=x^2y i+xy j+x^2z k
Partial derivatives computation are $\partial F_1/\partial x, \partial F_1/\partial y, \partial F_1/\partial z$ etc
- 4. 1. 0.
2. - 1. 0.
0. 12. 4.
div of F=-1

Find divergence of a vector function

$$\vec{F} = x^2 y \hat{i} + xy \hat{j} + y^2 z \hat{k}$$

$$\text{Solution: } F_x = 2xy \quad F_y = x \quad F_z = y^2$$

$$\nabla \vec{F} = 2xy + x + y^2$$

$$\text{At } (-1,2,3) \quad \nabla \vec{F} = -1$$

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Curl of vector function $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$,

$$\text{curl}\vec{F} = \nabla \times \vec{F} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\hat{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\hat{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\hat{k}$$

//Program to find curl

clc

function F=fun(x)

F(1)=x(1)^2*x(2)

F(2)=x(2)*x(1)

F(3)=x(2)^2*x(3)

endfunction

disp("Vector function F=x^2y i+xy j+x^2z k")

x=[-1,2,3]

f=numderivative(fun,x)

disp("Partial derivatives computation are $\partial F_1/\partial x, \partial F_1/\partial y, \partial F_1/\partial z$ etc")

disp(f)

//curl F is $(\partial F_3/\partial y - \partial F_2/\partial z)\hat{i} + (\partial F_1/\partial z - \partial F_3/\partial x)\hat{j} + (\partial F_2/\partial x - \partial F_1/\partial y)\hat{k}$

cf=[f(3,2)-f(2,3),f(1,3)-f(3,1),f(2,1)-f(1,2)]

disp("curl of F="+string(cf(1))+ "i"+string(cf(2))+ "j"+string(cf(3))+ "k")

OUTPUT

Vector function F=x^2y i+xy j+x^2z k

Partial derivatives computation are $\partial F_1/\partial x, \partial F_1/\partial y, \partial F_1/\partial z$ etc

- 4. 1. 0.

2. - 1. 0.

0. 12. 4.

curl of F=12i+0j+1k

Find curl of a vector function

$$\vec{F} = x^2y\hat{i} + xy\hat{j} + y^2z\hat{k}$$

Solution: $F_{1y} = x^2$ $F_{1z} = 0$

$$F_{2x} = y$$

$$F_{2z} = 0$$

$$F_{3x} = 0$$

$$F_{3y} = 2yz$$

$$\nabla \times \vec{F} = 2yz\hat{i} + 0\hat{j} + (y - x^2)\hat{k}$$

$$\text{At } (-1, 2, 3) \nabla \times \vec{F} = 12\hat{i} + 0\hat{j} + 1\hat{k}$$

****Additional examples**

LAB 1: 1) Find the last digit of the following

- i) 3^{200} ii) 7^{291} iii) 3^{50} iv) 9^{51} v) 3^{100} vi) 7^{100}

2) Find the remainder of the following

- i) 7^{100} when divided by 12
ii) 3^{100} when divided by 10
iii) 2^{300} when divided by 5
iv) 3^{200} when divided by 13
v) 5^{100} when divided by 6

LAB 2: Find the remainder of the following

- i). 2^{1000} when divided by 17
ii) 3^{50} when divided by 7
iii) 5^{200} when divided by 11
iv) 3^{50} when divided by 11
v) 3^{200} when divided by 13
v) 5^{225} when divided by 13

LAB 3: 1) Prove that $6!+1$ is divisible by 7

- 2) Prove that $12!+1$ is divisible by 13
3) Prove that $16!+1$ is divisible by 17
4) Prove that $2!+19$ is divisible by 21
5) Prove that $1!+5$ is divisible by 6
6) Prove that $4!+11$ is divisible by 35
7) Prove that $10!+111$ is divisible by 143

LAB 4: 1) Evaluate $\int \sin^6 x \, dx$

- 2) Evaluate $\int \sin^7 x \, dx$
3) Evaluate $\int \sin^5 x \, dx$

LAB 4: 1) Evaluate $\int \cos^6 x \, dx$

- 2) Evaluate $\int \cos^7 x \, dx$
3) Evaluate $\int \cos^5 x \, dx$

LAB 5: 1) Evaluate $\int_0^{\pi/2} \sin^2 x \, dx$

- 2) Evaluate $\int_0^{\pi/2} \sin^7 x \, dx$
3) Evaluate $\int_0^{\pi/2} \sin^4 x \, dx$

LAB 5: 1) Evaluate $\int_0^{\pi/2} \cos^4 x \, dx$

2) Evaluate $\int_0^{\pi/2} \cos^7 x \, dx$

3) Evaluate $\int_0^{\pi/2} \cos^3 x \, dx$

LAB 6: 1) Evaluate $\int \tan^3 x \, dx$

2) Evaluate $\int \tan^4 x \, dx$

3) Evaluate $\int \tan^7 x \, dx$

LAB 6: 1) Evaluate $\int \cot^5 x \, dx$

2) Evaluate $\int \cot^6 x \, dx$

3) Evaluate $\int \cot^3 x \, dx$

LAB 7: 1) Evaluate $\int \sec^4 x \, dx$

2) Evaluate $\int \sec^6 x \, dx$

3) Evaluate $\int \sec^3 x \, dx$

LAB 7: 1) Evaluate $\int \operatorname{cosec}^4 x \, dx$

2) Evaluate $\int \operatorname{cosec}^5 x \, dx$

3) Evaluate $\int \operatorname{cosec}^3 x \, dx$

LAB 8: 1) Evaluate $\int x^4 (\log x) \, dx$

2) Evaluate $\int x^2 (\log x)^2 \, dx$

3) Evaluate $\int x^3 (\log x)^4 \, dx$

LAB 8: 1) Evaluate $\int x^2 e^{5x} \, dx$

2) Evaluate $\int x^4 e^x \, dx$

3) Evaluate $\int x^5 e^{3x} \, dx$

LAB 9: 1) Evaluate $\int \sin^2 x \cos^4 x \, dx$

2) Evaluate $\int \sin^2 x \cos^2 x \, dx$

3) Evaluate $\int \sin^3 x \cos^4 x \, dx$

LAB 9: 1) Evaluate $\int_0^{\pi/2} \sin^2 x \cos^4 x \, dx$

2) Evaluate $\int_0^{\pi/2} \sin^2 x \cos^2 x \, dx$

3) Evaluate $\int_0^{\pi/2} \sin^3 x \cos^3 x \, dx$

LAB 10: 1) Verify Euler's Theorem for the function $u = \frac{x^2+y^2}{x+y}$

2) Verify Euler's Theorem for the function $u = \frac{x^3+y^3}{x+y}$

3) Verify Euler's Theorem for the function $u = \frac{x^3+y^3}{x-y}$

4) Verify Euler's Theorem for the function $u = (x^2 + y^2)^{1/2}$

5) Verify Euler's Theorem for the function $u = \frac{xy}{x+y}$

LAB 11: 1) Expand $f(x, y) = e^x \sin(y)$ by Maclaurin's theorem in powers of x & y up to 2nd degree

2) Expand $f(x, y) = \cos(x) \cos(y)$ by Maclaurin's theorem in powers of x & y up to 2nd degree

3) Expand $f(x, y) = \sin(x) \sin(y)$ by Maclaurin's theorem in powers of x & y up to 2nd degree.

4) Expand $f(x, y) = \sin(x+y)$ by Maclaurin's theorem in powers of x & y up to 2nd degree.

5) Expand $f(x, y) = \cos(x+y)$ by Maclaurin's theorem in powers of x & y up to 2nd degree.

LAB 12: 1) Find the gradient of a scalar function $F = x^2y + y^2x + z^2$

2) Find the gradient and of a scalar function $F = x^2 + y^2 + z^2$

3) Find divergence of a vector function $\vec{F} = xy^2i + 2x^2yzj + 3yz^2k$

4) Find divergence of a vector function

$$\vec{F} = (y^2 + z^2 - x^2)i + (z^2 + x^2 - y^2)j + (x^2 + y^2 - z^2)k$$

5) Find curl of a vector function $\vec{F} = xy^2i + 2x^2yzj + 3yz^2k$

6) Find curl of a vector function

$$\vec{F} = (y^2 + z^2 - x^2)i + (z^2 + x^2 - y^2)j + (x^2 + y^2 - z^2)k$$